

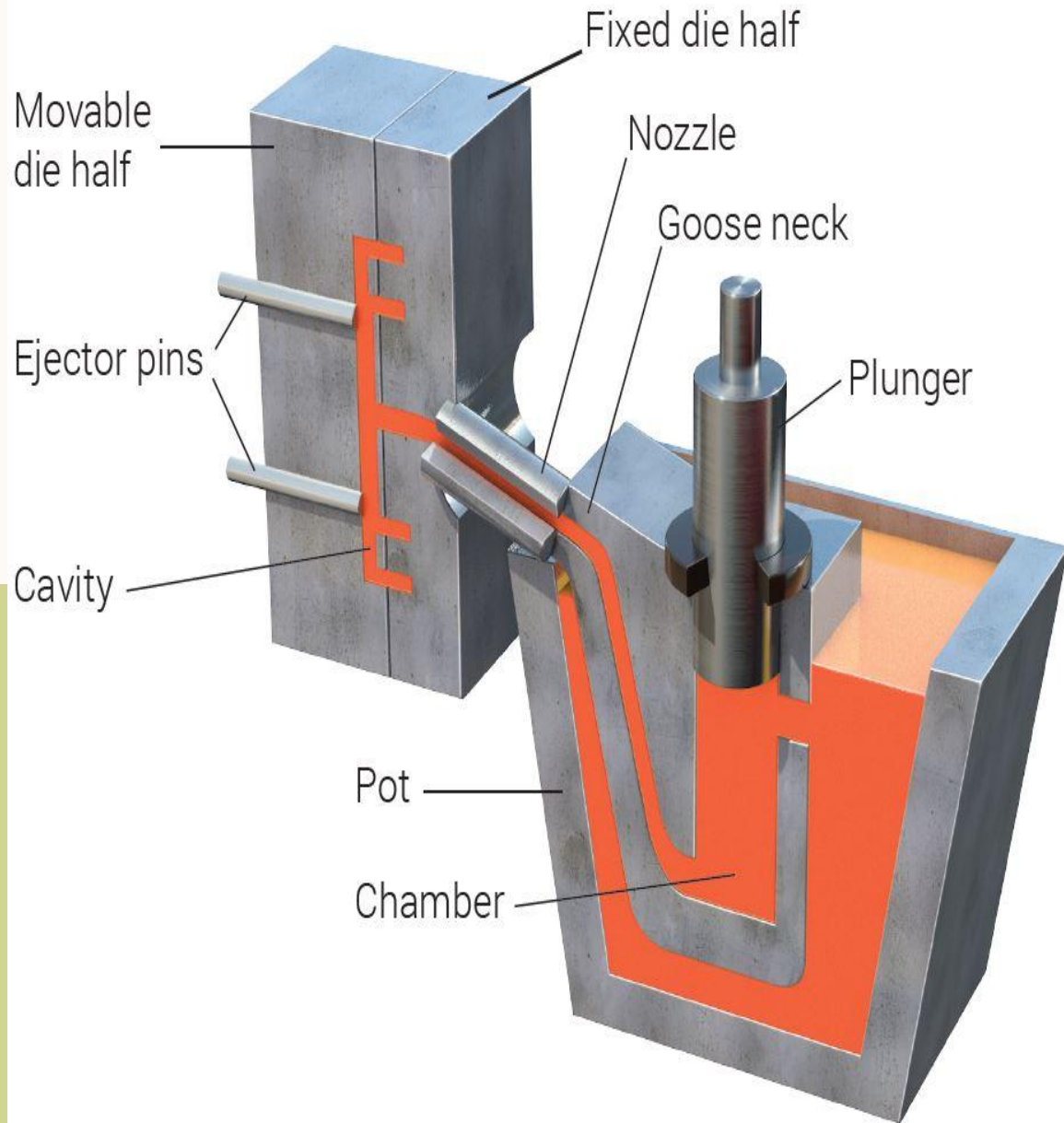
MARUT DRONES TECHNOLOGY PS KRITI 2026

System feasibility &
readiness for drone
integration.

CONTENTS

1. Manufacturing ease
2. Robust electronics
3. Simulations
4. Analysis
5. Industry readiness
6. Mechanical simplicity
7. Drone Integration

An overview of system feasibility and operational readiness for drone integration.



MANUFACTURING EASE

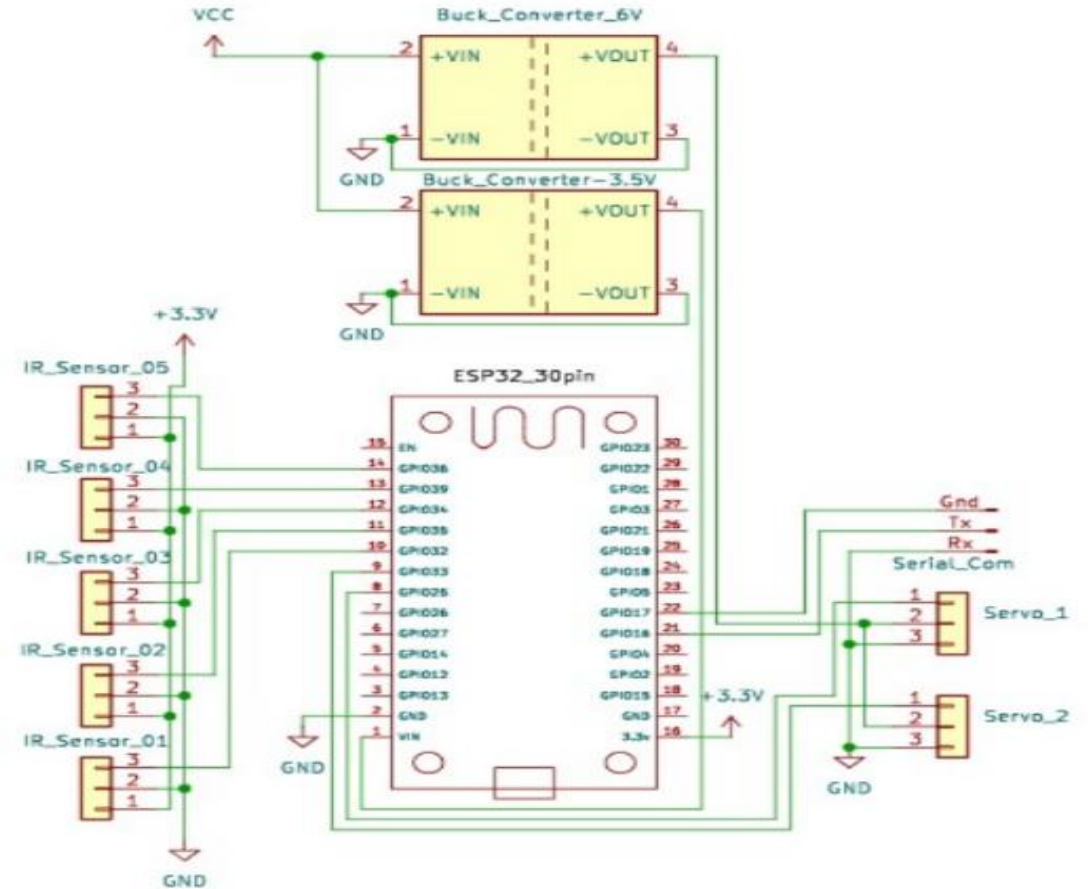
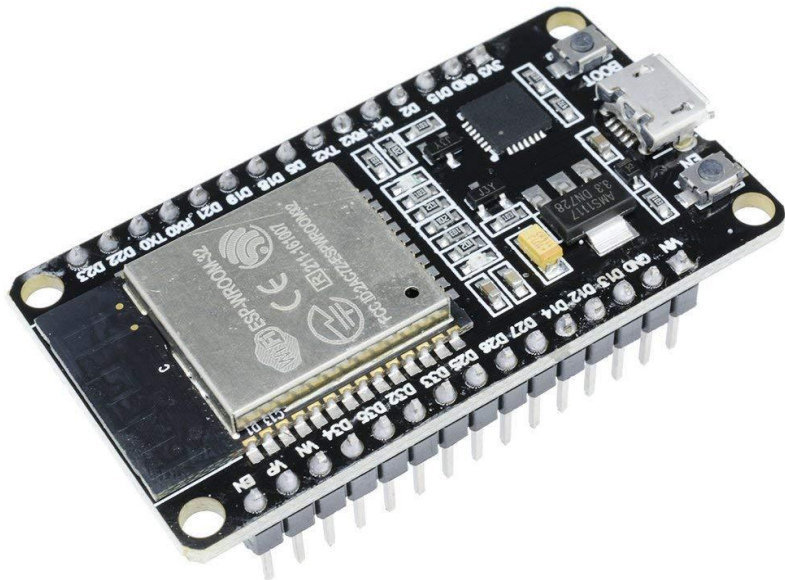
High-performance components designed for maximum durability and environmental resilience.

DESIGNED TO BE MANUFACTURED

- Die-casted in durable 4XX.X series Aluminum due to it's high castability.
- Composed of Aluminum-Silicon alloy for reduced wear and corrosion.
- Ensures that the tolerances are very low.
- Welding avoided to maximum extent.
- Only welded at 5 Interfaces.
- Used very few sharp edges for the hopper design.

ROBUST ELECTRONICS

High-performance components designed for maximum durability and environmental resilience.

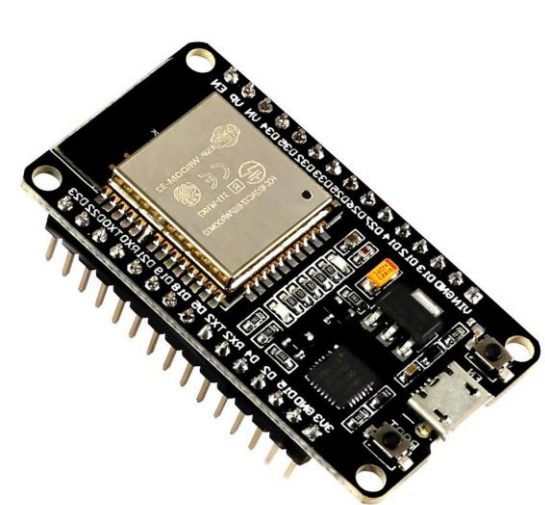


ROBUST ELECTRONICS

- Low current drawn and at 12 Volts. Works at very low power.
- No usage of Arduino. ESP used instead. (dual core design is practical and less prone to failure, also it has wi-fi capability)
- STM to be used in the future.
- IR sensors used instead of ultrasonic sensors
- MG995 full metal gear servo used



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SENSORS AND SERVO FUNCTIONALITY

Infrared over Ultrasonic

1. Ultrasonic sensor affected adversely by sound and vibrations.
2. IR sensors work in a closed system and thus work better.
3. Faster response for precise dispensing control.
4. Zero dead zones for high-precision, close-range detection.

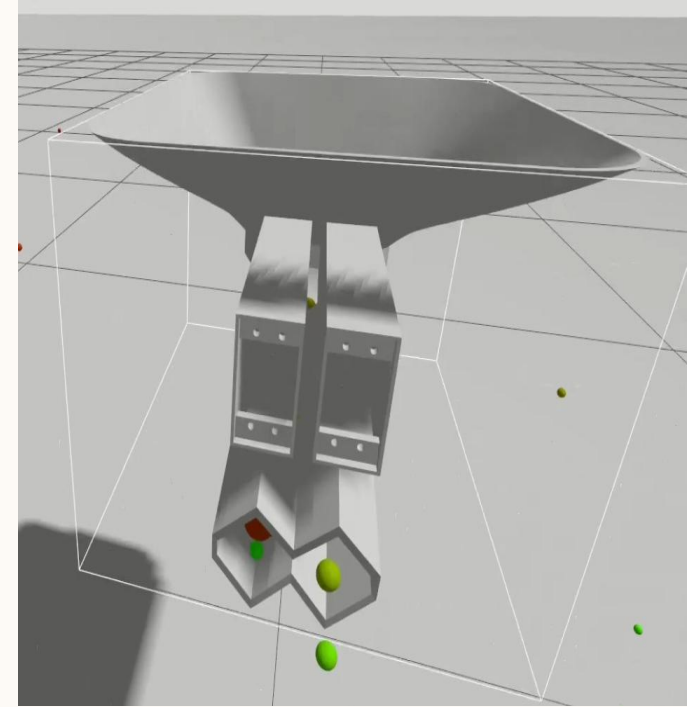
Mg995R Servo

1. It is a full metal gear servo and is thus much more durable than plastic
2. Functionality of the vibrator also taken over by the servo.
3. The amplitude of the vibration can be controlled and the number of components does not increase (thus also reduced number of risk mode).



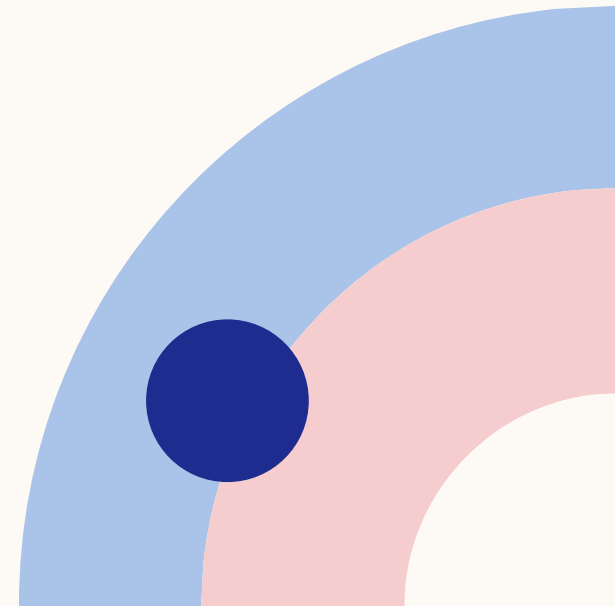
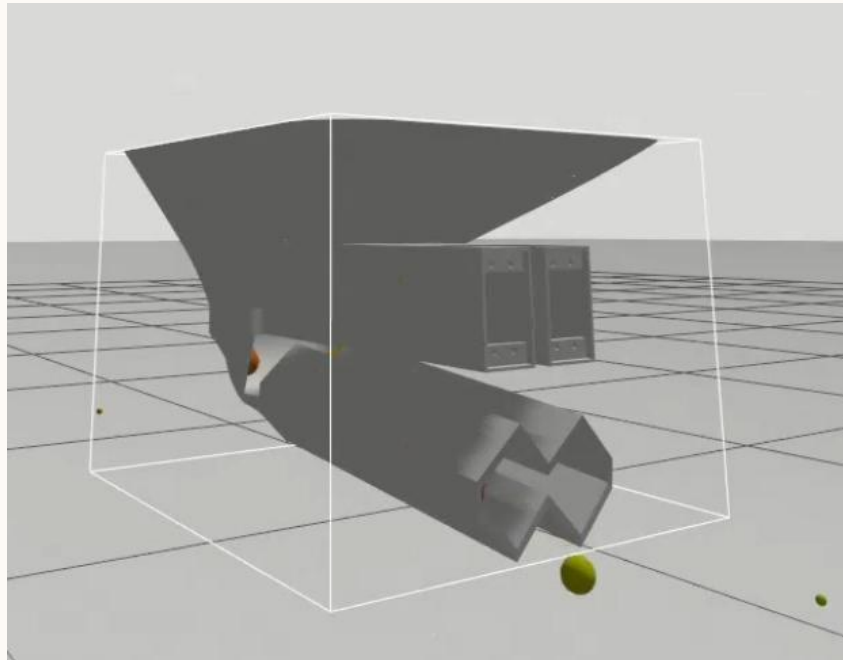
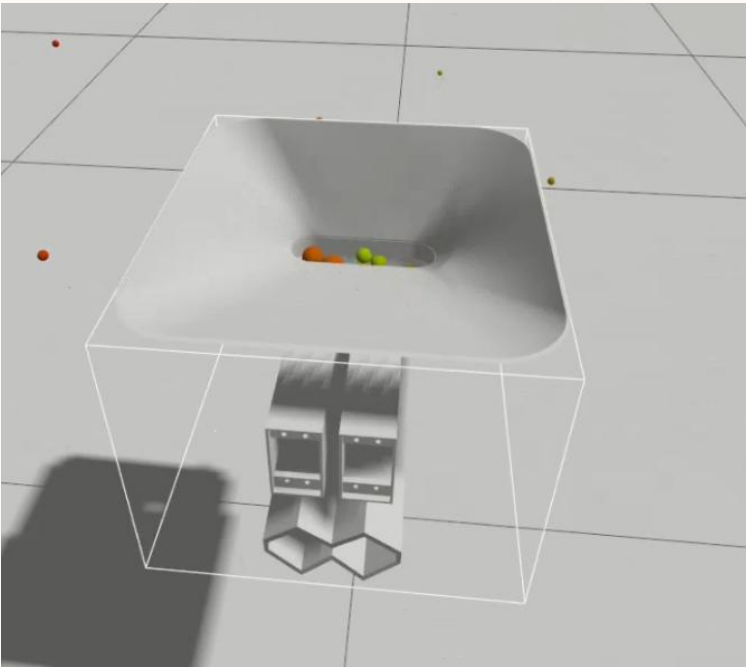
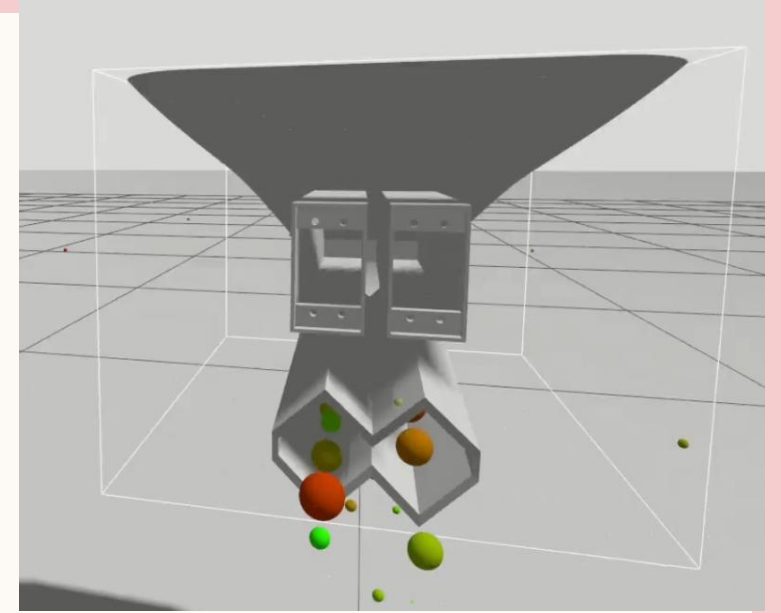
SIMULATIONS

All performance and operational simulations were conducted using industry-standard software to ensure realistic and reliable results.



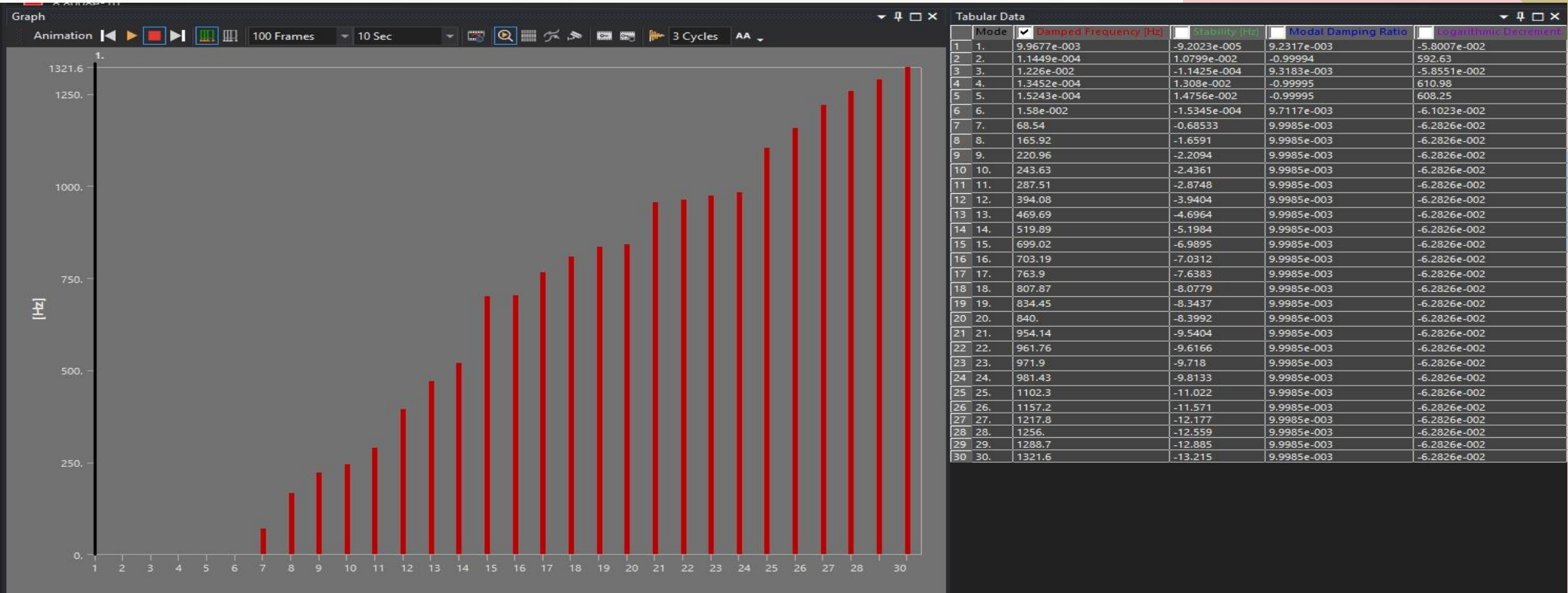
DETAILS ON SIMULATIONS

1. All simulations and tests have taken place on industry-level software: Gazebo.
2. All friction, air resistance and any other such resistive forces taken into account for simulations.
3. Real life conditions were simulated as close as possible.



ANALYSIS

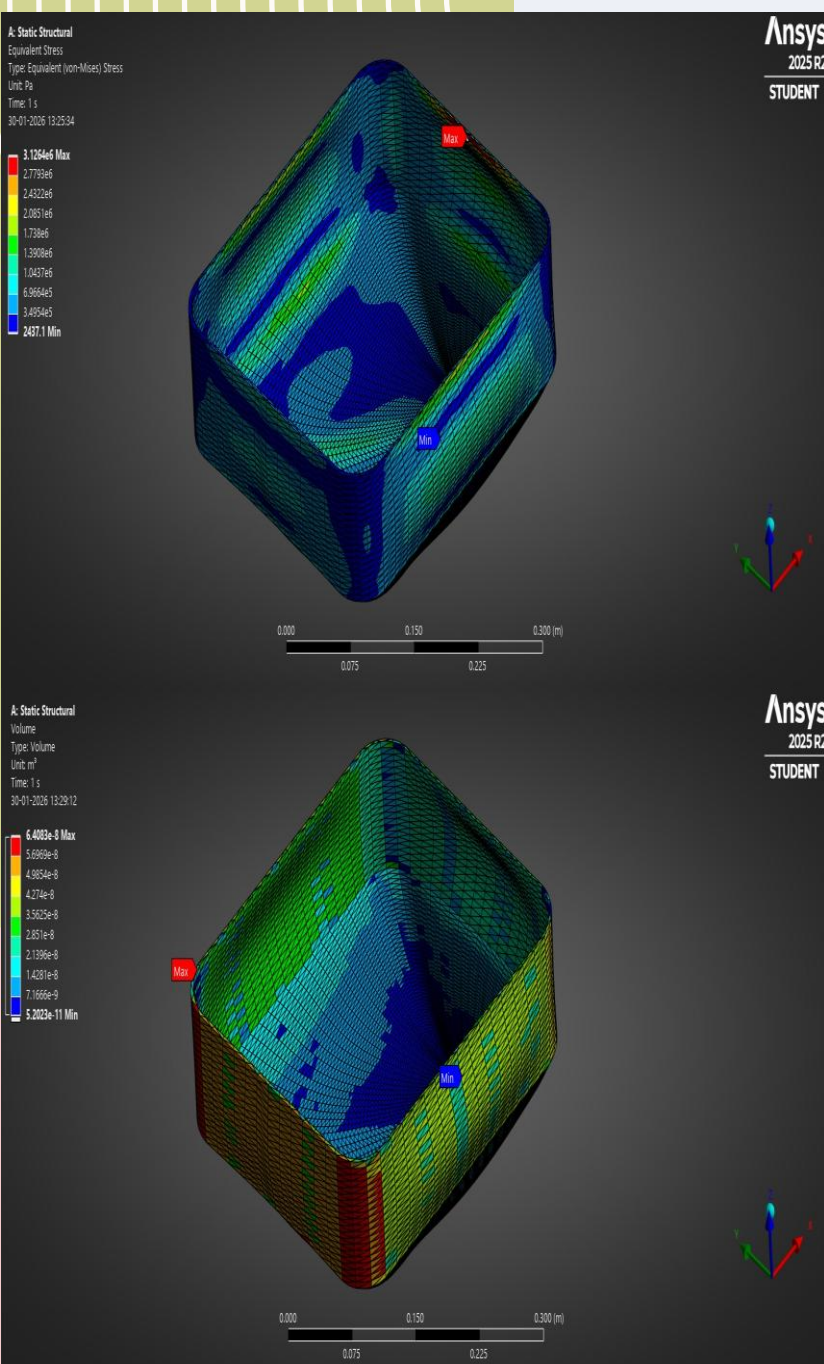
ANSYS-based FEA/FEM simulations validate the hopper's structural strength, stability, and safe performance under real operational loads.



DETAILS ON ANALYSIS

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- ANSYS FEA analysis shows the hopper can safely withstand operational loads without breaking.
- Stress and deformation values are within permissible material limits.
- Factor of safety obtained confirms strong and reliable structural design.
- FEM simulation validates stability under vibration and real drone conditions.
- Overall analysis proves the mechanism is robust and ready for practical use.



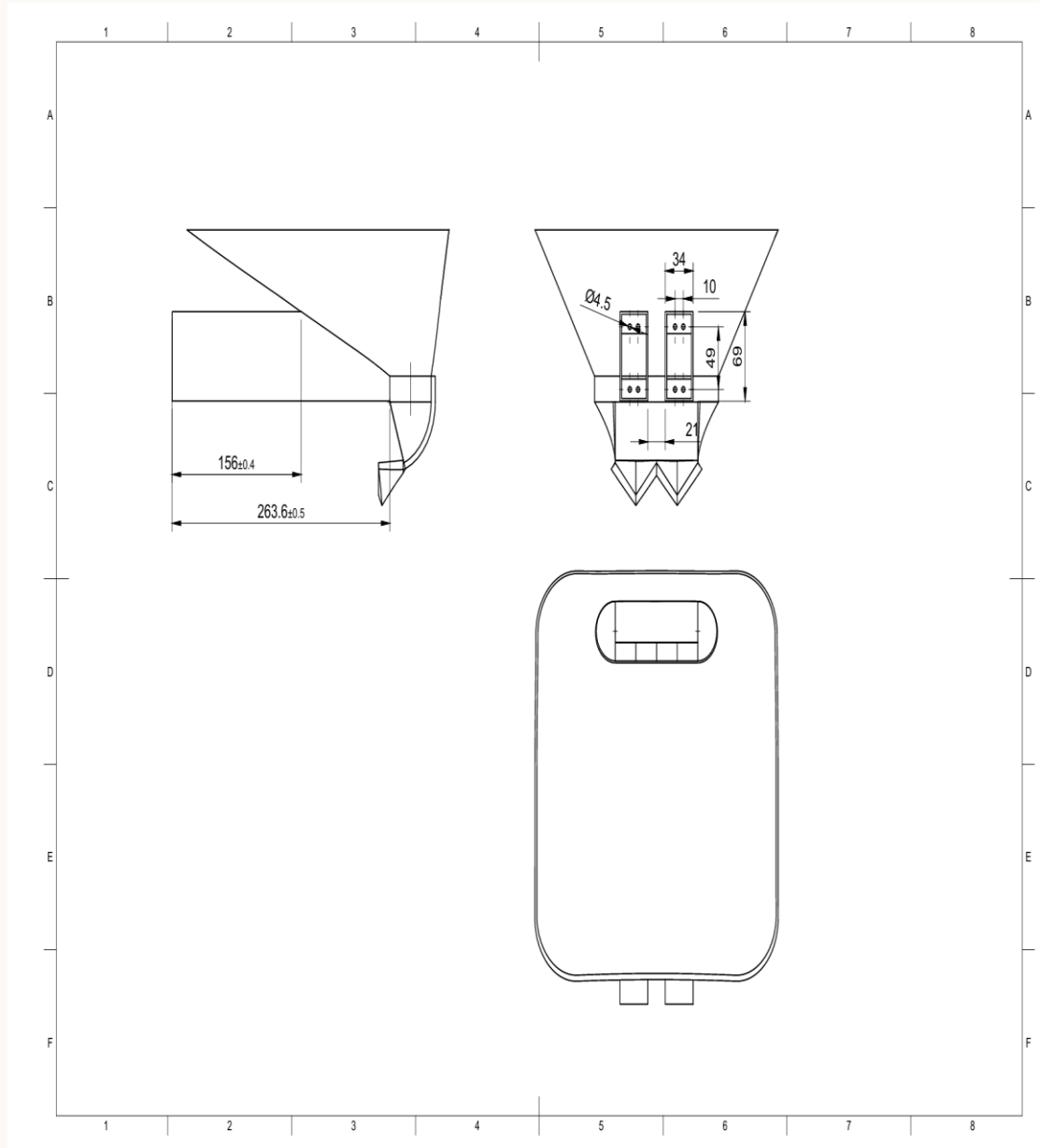


INDUSTRY READINESS

AN INDUSTRY-STANDARD DESIGN

- Suitable for real outdoor and agricultural environments.
- Compact and lightweight structure due to usage of Aluminum.
- Tested for smooth flow and anti jamming performance.
- Operated effectively under vibration conditions.
- Easily mountable on the given drone platform using a simple mounting procedure with nuts and bolts.



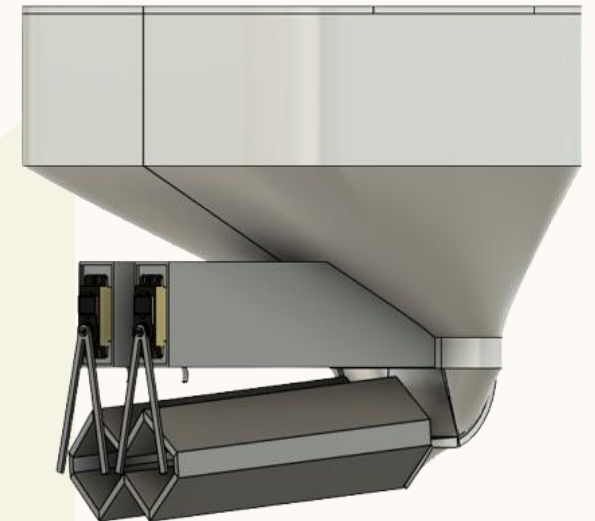


MECHANICAL SIMPLICITY

A robust yet simple mechanical design ensures durability, reliability, and efficient performance in rough terrains.

ROBUST MECHANICAL DESIGN

- Stress concentration avoided due to our chosen design principle.
- The V-file structure and servo actuation design work in tandem to produce a reliable and practical design.
- Scalable for commercial deployment with different dimensions, with the core design principal remaining the same.
- Suitable for large-scale afforestation and precision agricultural missions



**THANK
YOU**